

A SYSTEMS ENGINEERING APPROACH TO APPRAISE CYBERSECURITY RISKS OF CNS/ATM AND AVIONICS SYSTEMS

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Abstract

The aggregate use of complex cyber-physical systems, incorporation of artificial intelligence and autonomy in flight operations has reconceptualised the aerospace cybersecurity paradigm to a new era. Cybersecurity is about the prevention or mitigation of malicious acts undertaken to either compromise systems directly or access valuable information that could be subject to criminal misuse. The cybersecurity of Communication, Navigation, and surveillance / Air Traffic Management (CNS/ATM) systems is aimed at controlling the effects of such threats and enhancing the safety of flying passengers. Besides, the cybers resilience of state-of-the-art CNS/ATM and avionics systems have been impacted by convoluted risk management issues. The security of legacy aviation and airport infrastructure has been curtailed owing to their annexure to public networks, which constitute a significant risk to the aviation industry and the general public as a whole.

The existing control mechanisms in place, with the aim of assuring Confidentiality, Integrity, and Availability (CIA) of information, have a multitude of deficiencies in deciding the extent of vulnerability of systems, the severity of impact and subsequently overall risk rating. Weaknesses in some of the security controls which are intended to mitigate the impact of cyber-attacks also contribute towards this issue. A risk-based system engineering solution is envisioned in managing today's cyber-physical systems which may comprise of interwound Systems of Systems (SoS). This paper aims to investigate threats to aviation assets, vulnerabilities of aviation systems and to adopt a holistic approach to mitigate possible risks. The article also recommends a novel methodology to manage risks by enhancing the consistency and accuracy of risk attributes of CNS/ATM systems, avionics and airport systems.

Introduction

Increasingly more threat actors, including state level and organised cyber-criminal enterprises, are engaged in developing enhanced offensive resources or acquiring them through various sources for illicit and malicious activities. A new generation of targeted cyber-attacks called Advanced Persistent Threat (APT) have evolved, and they are highly sophisticated and deploy many intrusion techniques to evade detection and silently extract information. They are aimed to attack a specific system with the intention of compromising them and often gain access to information by encrypting and compressing it to bypass firewalls and intrusion detection systems. A blended attack is a sequence of sophisticated cyber-attacks which aims to bewilder network and security administrators by launching one kind of apparent attack while simultaneously executing a second attack that is obscured by the first attack and infiltrate organisations' IT infrastructure to acquire critical information or to manipulate them. In this backdrop, assessing aviation security threats and vulnerabilities has become a challenging task owing to data complexity and uncertainty of critical risk attributes such as the likelihood of occurrence and impact level. The scope of this paper is to investigate aviation system threats and vulnerabilities and to present a cybersecurity risk assessment methodology using Analytical Hierarchy Process (AHP) and Fuzzy Inference System (FIS).

The aviation system architecture, as depicted in Figure 1, is composed of five major domains: (1) Aircraft, (2) ATM system, (3) Airport and Airline network[1,2], (4) System Wide Information Management (SWIM), and (5) Ground radio navigation and landing aids.

flight crew (Figure 2). This approach has enabled least amount of crew engagement, and the role of crew members is sometimes limited to the supervision and monitoring of system functions. Consequently, the aerospace industry has gained both economic and operational benefits with the

implementation of IT and a multitude of wireless communication technology [4].

The potential threats and vulnerabilities of aircraft system and associated IT infrastructure are summarised below.

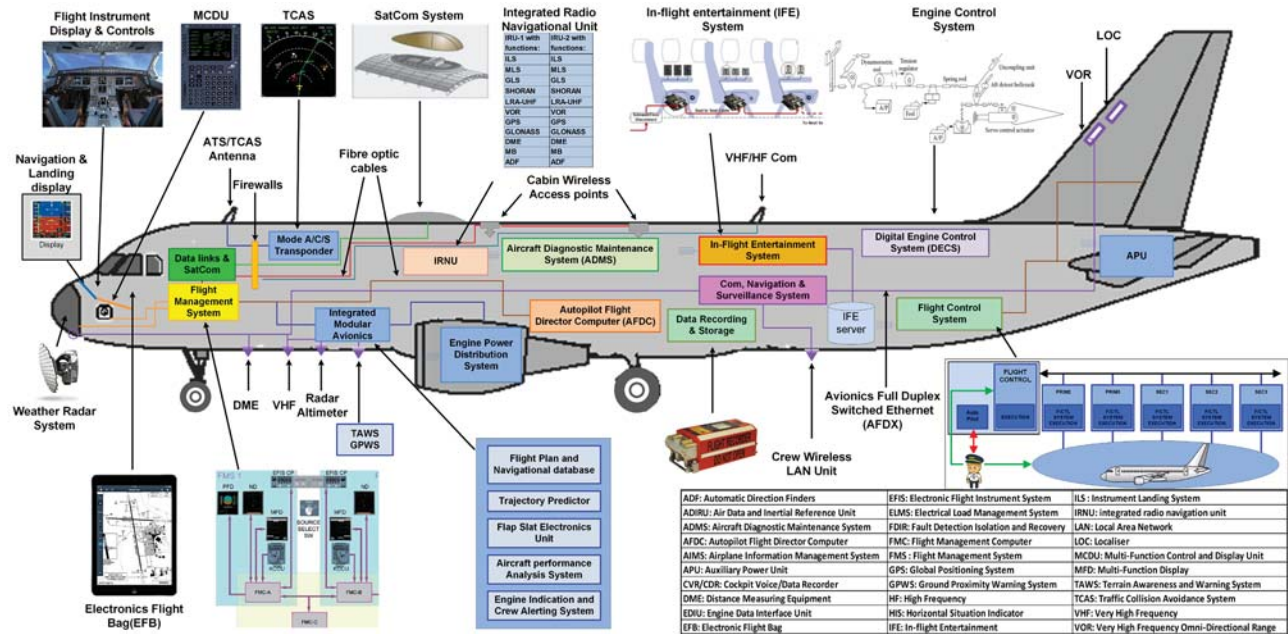


Figure 2. Avionics, Aircraft Information Processing Systems and Interconnections

- Zero-day attack to FMS: Attackers exploit software vulnerability and manipulate flight plan data, coordinates of trajectories, navigational aid and weather data
- Possible lack of network segregations between flights Management System (FMS), In-Flight Entertainment (IFE) system and passenger wireless network on some aircraft types
- Malware, virus or worm attack to FMS:
- ADS-B flight ID spoofing (Altered Aircraft ID)
- Malware attack on the ACARS system. Controller Pilot Data Link Communication [CPDLC] message deletion or alteration compromising route, terminal, aeronautical and weather information
- Malware (or Stuxnet) attack on aircraft radar transponder (modified Aircraft ID and altitude)
- Global Navigation Satellite System (GNSS) Jamming and spoofing (Malware)
- Unauthorised access to Electronics Flight Bag (EFB) and Navigational-aid databases
- Use of proprietary encryption key algorithms between avionics and ground control systems for data communication
- Compromised private keys being used between avionics and ground control systems when using Public Key Infrastructure (PKI) for data communication
- Denial-of-Service (DoS) attack on aircraft datalink application
- Privilege escalation of ADS-C and CPDLC logon credentials
- Country-specific legislations require disclosure of Avionic data link secret key and decryption algorithm to country's authority (when cryptosystems are being used in data communication)

- Use of Industrial, Scientific, and Medical (ISM) radio band (802.11 b/g) to update aircraft system components (AHM, LSAP, VDar, EFB) using Boeing GateLink Terminal Cellular Unit (TCU)
- Use of unsecured (or unencrypted) wireless dataloaders to update aircraft information databases and configuration
- In-flight data update during flight using the Internet via a satellite link with no SSL/TSL or VPN connection
- No active intrusion detection and/or active response mechanism (such as IPS & firewall) to cyber-attack, and

- Regulation restrictions – delay in important security updates).
- Use of common default engineering login (username and password)

Air Traffic Management - Threats and Vulnerabilities

Air traffic controllers predominantly monitor position, speed, and altitude of flights within the designated airspace. Air traffic surveillance systems (Radar, ADS-B and ADS-C) are used for the purpose of providing aircraft with information for significant deviations from the nominal flight plan by means of voice and data communication (see Figure 3).

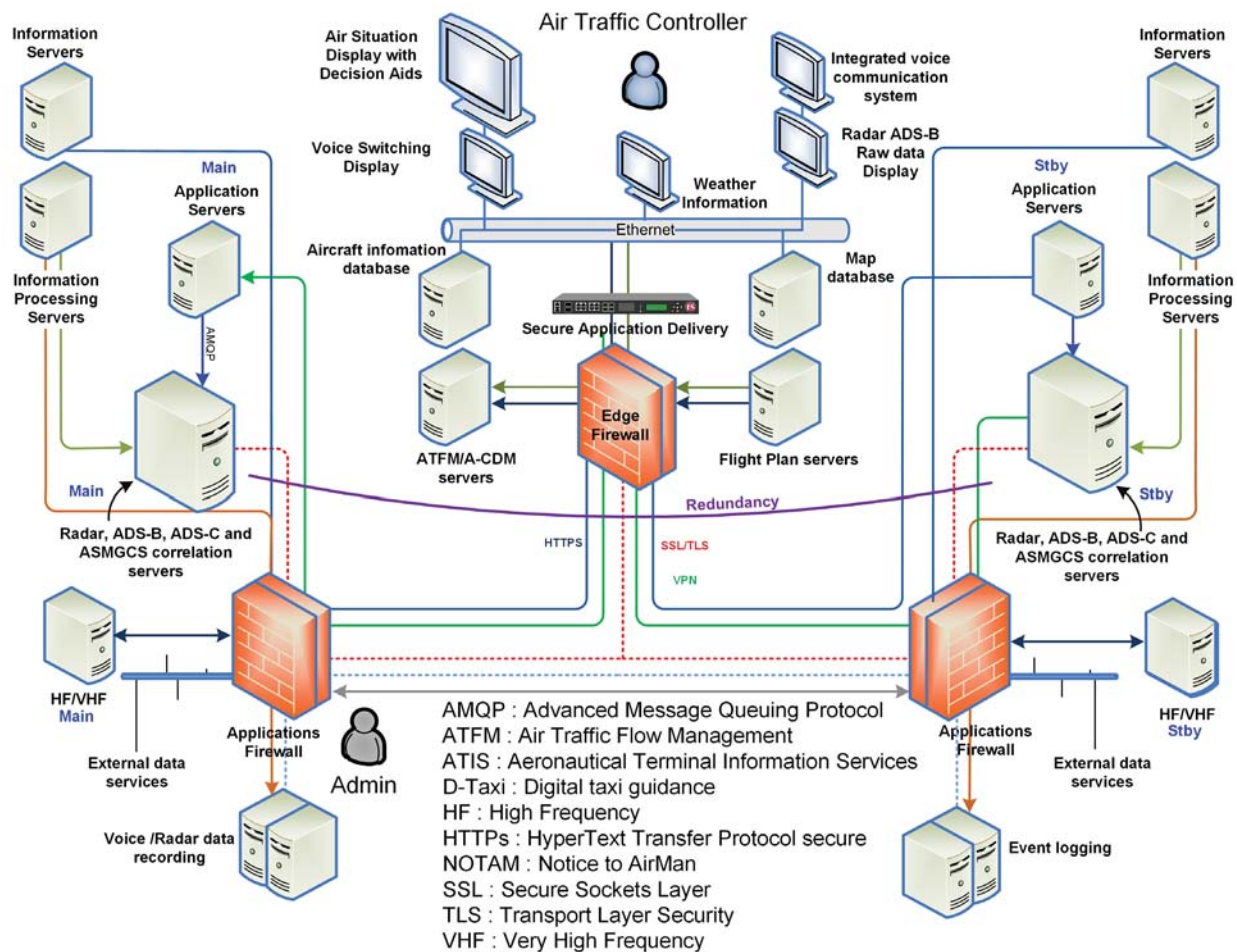


Figure 3. A Typical Configuration of an ATM System

The ATM mission support systems are based on a distributed architecture implemented over a wide area network between main centers and neighboring Flight Information Regions (FIR).

Each ATM component is based on the same basic architecture and includes the same core systems supplemented by additional sub systems. The voice and data are exchanged between external

communication systems via various interfaces and in different data formats. The primary operational functions provided by an air traffic mission system are:

- Surveillance information on situational air displays
- Flight data management
- Coordination functions between adjacent sectors
- Conflict detections
- Arrival / Departure management
- Aeronautical data management
- Workload management function
- Ground facilities management, and
- Voice switching for Air/Ground and Ground/Ground Communication.

The following vulnerabilities could potentially exist in ATM system that could be affected by security threats depending on system configuration:

- Inadequate network segregation between ATM and corporate/public systems
- Man-in-the-middle (MITM) + replay (ADS-B) attack where altered ADS-B aircraft location data is fed to ATM system
- Information leak through covert channel (storage or timing) attack
- Unsecured (or unencrypted) data and voice (VoIP) communication over public shared network between ATM centres
- Inadequate network perimeter and endpoint security controls to inspect and prevent encrypted malware
- SQL injection (flight plan, aircraft parameter, route, reporting points, weather, and aeronautical information databases)
- Privilege escalation attack on ATM system
- Inadequate application and endpoint security protection (virus, worms, Trojan, rootkit, spyware, blended threat, and adware)
- Remote access session hijacking (Admins use remote VPN to troubleshoot ATM systems and networks)

- Privilege escalation to ACARS system (ADS-C and CPDLC messages)
- No software patch management for ATM systems (leading to Zero-day attack)
- No Disaster Recovery Plan (DRP) or Business Continuity Plan (BCP) for ATM mission support systems
- No Security Information and Event Management (SIEM) mechanism for ATM system and network admin operations
- Privilege escalation to Notice to Airmen (NOTAM) system
- Unauthorised interception or interference to wireless ATC data communication
- DoS attack on core/distribution network connecting adjacent ATSP facilities
- An attack on symmetric key (Known plaintext, Chosen plaintext or Chosen ciphertext) between avionics and ATM system when using encrypted data communication
- Inadequate ATM system physical security and access controls
- No patching on GPSRAIM servers, and
- No ATM incident response plan for possible cyber-attacks.

Airport, Airline and ANSP Network - Information security Risk Assessment

The communication systems that are currently in operation for airport voice and data communication have reached their limits and is characterised by many custom legacy telecommunication protocols, each with their self-contained information systems.

The following threats and vulnerabilities could exist in an airport and airline communication network depending on the type of systems in operation and their configurations (see Figure 4).

- Inadequate network segregation between airport and corporate/public systems
- Spear phishing attack. Targets key executives/administrators and attack their computers with malware
- Ransomware attack targeting executives, administrators and key employees

- Highly skilled state sponsored reconnaissance attack (Attackers with a large amount of resources at their disposal carry out this type of attacks)
- Trojan, fishing, spam and social engineering attack on airport and airline IT infrastructure
- Use of internal infectious sources (PCs) or networks as emitters to conduct botnet attacks
- Unsecured physical access to IT infrastructure
- Use of legacy ICS on airport infrastructure with remote monitoring via the Internet
- Integration of less secure legacy technology systems with secure systems
- Employees or contractors misuse restricted information either by accidentally or deliberately (e.g., when threatened by terrorists)
- Use of shared login in aviation and airport systems (e.g., airline ticketing, ramp handling and maintenance/control system)
- Use of uncertified/unaccredited commercially off the shelf (COTS) devices in airport systems
- Distributed Denial-of-Service (DDoS) (as a botnet attack) using malware infected system
- Use of default passwords in airport systems, remote and maintenance web consoles (Unauthorised use of airport IT infrastructure)
- Advanced persistent threat (APT)
- Use of open communication and legacy protocols for remote access and control
- Use of unencrypted communication for ACARS and ATN (satellite and/or ground-based) system
- Inadequate security configuration for LAN, WAN, IT Equipment Hardware & Software
- Unsecured physical network access to arrival/ departure control, ramp, fuel, catering and maintenance systems

- SQL injection to Flight Information Display System (FIDS)
- Excessive admin privileges or abuse of database privilege to staff authentication system (Biometric identification system)
- Malware attack on freight/ cargo /mail management system (weight and cargo screening)
- Open physical network and communication ports on critical infrastructure
- Use of Supervisory Control And Data Acquisition (SCADA) without adequate security controls
- Fraudulent deployment of hardware/ malware externally deployed communication infrastructure
- Radiation disturbance (electromagnetic and thermal) and performing DoS attacks
- Uncontrolled downloading and use of software (no software white-listing or black-listing), and
- Zero-day attack
- Poor supply chain security.

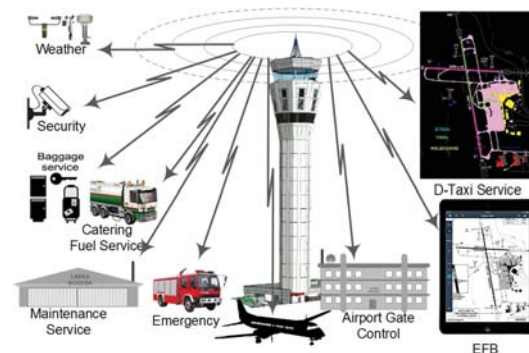


Figure 4. Airport and Airline Network

The proposed Aeronautical Mobile Airport Communication System (AeroMACS), which will become a predominant part of next-generation data link, allows airlines, ANSPs, aircraft and airport authorities to stay connected at airports and to discharge various communication operations. The new system will provide enhanced security and facilities management and maintenance supports in airfields [5]. Some of the vulnerabilities outlined in this paper would apply to AeroMACS too.

SWIM - Information Security Risk Assessment

The ICAO Manual on SWIM sets out the standards, infrastructure, and governance enabling the management of aeronautical information and its exchange between authorised aviation stakeholders [6] (see Figure 5). The primary objective of SWIM is to provide ATM information to users in an

interoperable manner. These services include flight data, weather updates, aeronautical data, and the following information governance functionalities.

- 1) Performance Improvement through the application of SWIM, and
- 2) Enabling airborne system participation in collaborative ATM through SWIM.

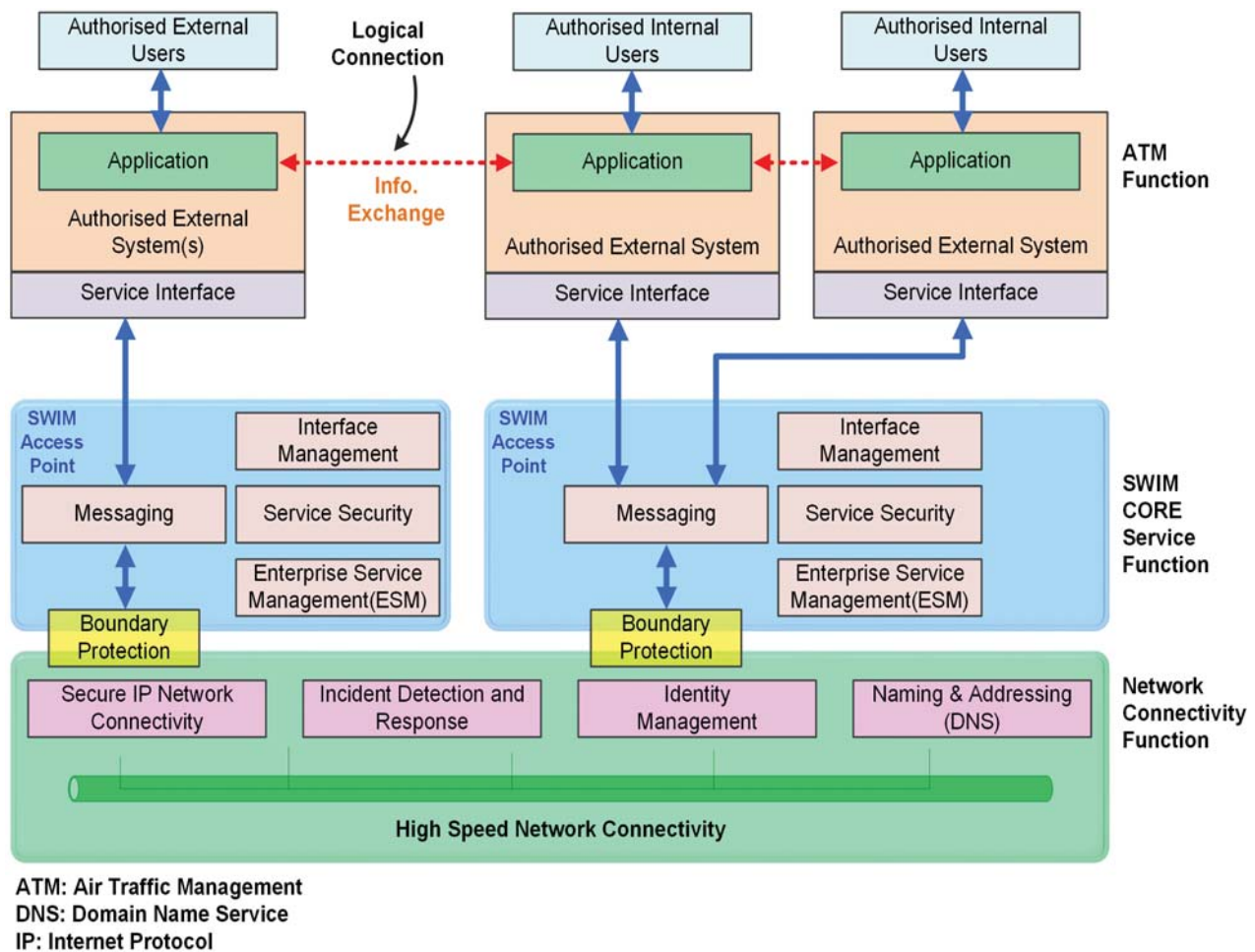


Figure 5. SWIM Architecture [6]

The SWIM concept introduces a noteworthy change in the practices of managing information during the total life cycle of the ATM process. The information is expected to be of high quality, delivered at the right time and disseminated to the right stakeholder, hence enabling net-centric dynamic ATM operations. The following potential security threats and vulnerabilities could be present in the SWIM.

- Inadequate security policies and procedures
- No data classification
- No authentication
- Loss of public Key Infrastructure (PKI) security credentials
- Network attack on SWIM (malware or injection of falsified information)

- Blended attack (consists of several attacks with one being obvious)
- Data encryption mechanism or secret key being used is compromised
- No software patch management
- No physical security control to SWIM hardware, servers and infrastructure
- No intrusion-prevention systems (IPS) (preferably at the edge of the network after firewall)
- No firewalls or intrusion detection system installed to protect SWIM network elements
- DoS attack on SWIM network connecting ANSP facilities.
- Poor access management (no principle of least privilege)
- Bring Your Own Device (BYOD) with lack of user access control
- SQL injection to a flight plan, trajectory, weather, and aeronautical information databases
- Inadequate or no Security Information and Event Management (SIEM); No traceability and forensics.
- Poorly designed SWIM web interfaces (with XSS vulnerability)
- Poor password policy
- No Data Loss Prevention (DLP) mechanism
- Tunneling: where one protocol is used to carry traffic for another protocol and this technique is used to access restricted information undetected

- Pivoting: A technique used to burrow deeper into a network by routing through one machine to another
- Exfiltration: Various ways to get information out of a network when it's been collected, and
- Poor supply chain security.

Ground Radio Navigation and Landing Aids

The use of the Internet-of-Things (IoT) infrastructure on legacy airport systems (Figure 6) has created a significant security risk and threat landscape. The identified threats and vulnerabilities for ground based navigational systems are summarised below.

- Malware infection of landing aids and navigational systems (such as ILS, DVOR, DME, GBAS, ASMGCS, SMR and AWOS)
- Use of unencrypted communication for control and monitor operation of airport infrastructure
- Use of legacy ICS, PLC and SCADA devices on airport infrastructure
- No software or firmware updates. The vulnerabilities remain unaddressed during the period of equipment lifecycle
- Intentional or unintentional damage to ground antenna and ancillaries causing denial of services (DoS)
- Unauthorised people from having access to ground facilities or equipment
- Malware infection of landing aids and navigational systems (such as ILS, DVOR, DME, GBAS, ASMGCS, SMR, and AWOS)
- Use of unencrypted communication for control and monitor operation of airport infrastructure

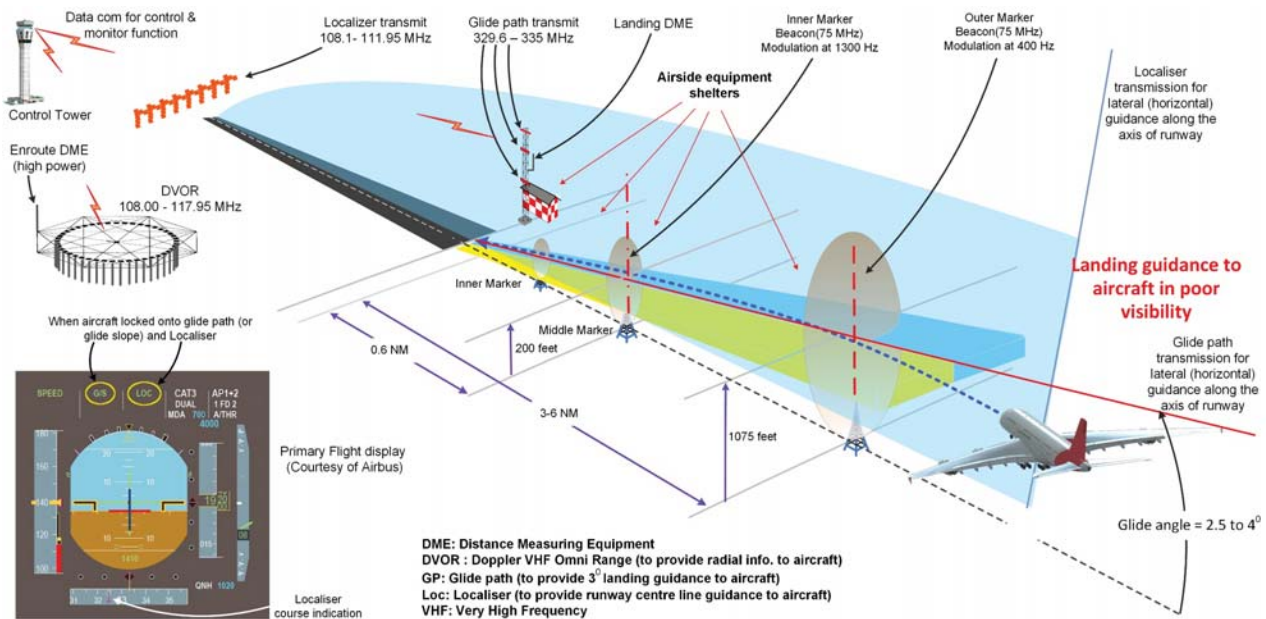


Figure 6. Ground Radio Navigation and Landing Aids

The Proposed Risk Assessment Methodology

A comprehensive security risk assessment should be performed at least once every two years to delve into the risks associated with the organisation's information and associated network infrastructure [7]. A corporate security risk assessment can only provide a snapshot of the risks at a particular point in time [8]. Also, the numeric weight assigns to 'likelihood,' and 'severity' criterion hugely depends on the expertise or the experience the assessor possesses on each area of the risk. The lack of knowledge or confusion of systems functions, interfaces, interconnections and interaction with other systems will lead to uncertainty in the judgment of risks. This situation will sometimes cause organisations to set aside high investment in strict security controls on less critical assets while deploying weak prevention measures on high-risk assets.

The risk assessment process takes off with a problem statement. A problem statement is a clear, concise definition of the security requirement for each system and the issues that need to be addressed by a risk assessment team. The statement may further set out deterministic needs for security, safety, integrity, confidentiality, and availability

along with criteria referring to the likelihood of occurrence of serious security breach/events and the possible consequences. This information can be derived from civil aviation security risk Common Body of Knowledge (CBoK) or using a mind map. Additional guidance and specifications on aerospace security issues are described in international regulations apply to the systems. e.g., the International Civil Aviation Organization (ICAO) and Federal Aviation Administration (FAA).

The next step is to analyse and organise the information using a structured technique based on mathematics and psychology. The consistency of the risk attributes is vital when performing the assessment. The Analytical Hierarchy Process (AHP) has different theories for prioritisation, consistency evaluation, and improvement and therefore it is chosen for the purpose.

Next, the threats, likelihood and vulnerability attributes are subjected to the defuzzifying process to obtain a definitive interpretation of risk level before the application of AS/ ISO/ IEC 27K guidelines as shown in Figure 7.

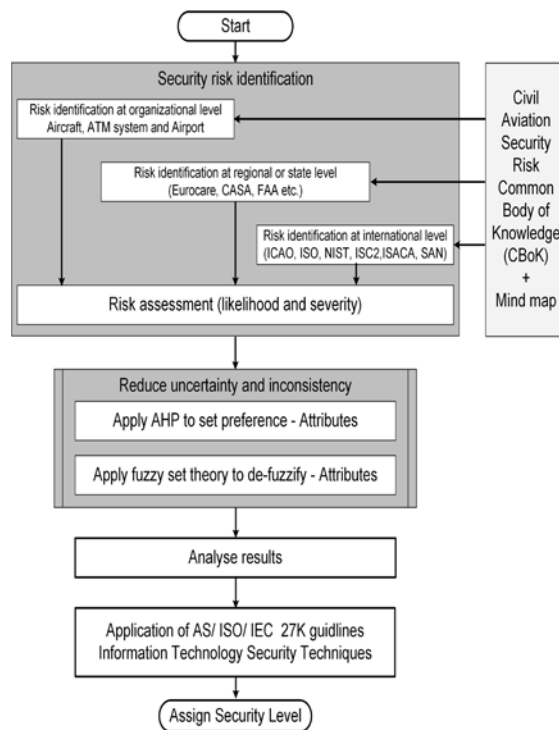


Figure 7. The Proposed Security Risk Assessment Methodology for Civil Aviation Systems

Analytical Hierarchy Process (AHP)

The AHP is a multi-criteria decision-making process that is used to set preferences among different attributes and a method to obtain ratio scales from paired comparisons [9]. The process is most useful where teams of security professionals, experts and assessors are working on a complex system assessment, especially those with high risks, involving human judgments and perceptions have long-term consequences and stakes. The AHP can be employed to prioritise essential elements or attributes of decisions which are difficult to quantify or compare under normal circumstances [10]. The outcome of the process is significantly accurate and consistent when the expertise and proficiency of the team members are impeded by their unique specialisations, interpretations, or perspectives.

The first step in the process is to decompose and structure attributes into a hierarchy of cyber-attacks and line up them against multiple aviation systems to be compared against (Figure 8). The input values are acquired from cybersecurity threat assessments (such as Injection and DoS) and level of importance of system in use (such as ATM system and Surveillance data system). Predominantly, the decisions are made from subjective opinions such as satisfaction, feelings, and preference of the security assessors.

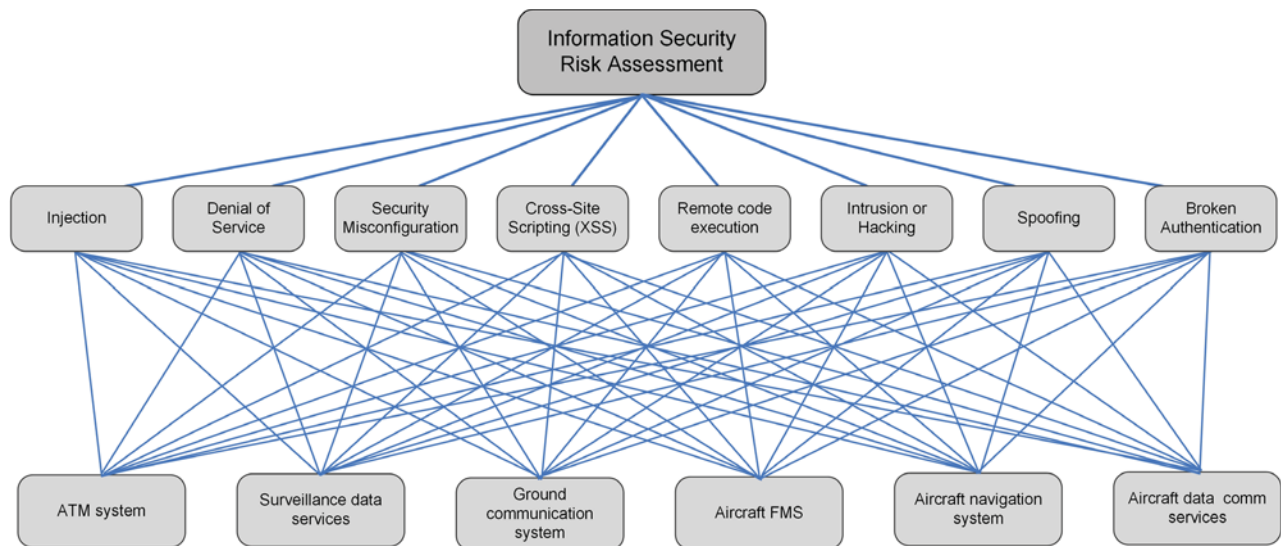


Figure 8. Pairwise Assessment of Security Risk using AHP

As depicted in Figure 8 there are 28 pairwise comparisons demonstrated by six critical aviation systems and eight significant attack types. (Injection, DoS, Misconfiguration, Remote code execution, Intrusion or Hacking, Spoofing, Broken Authentication, and XSS). The AHP steps are depicted in Figure 9

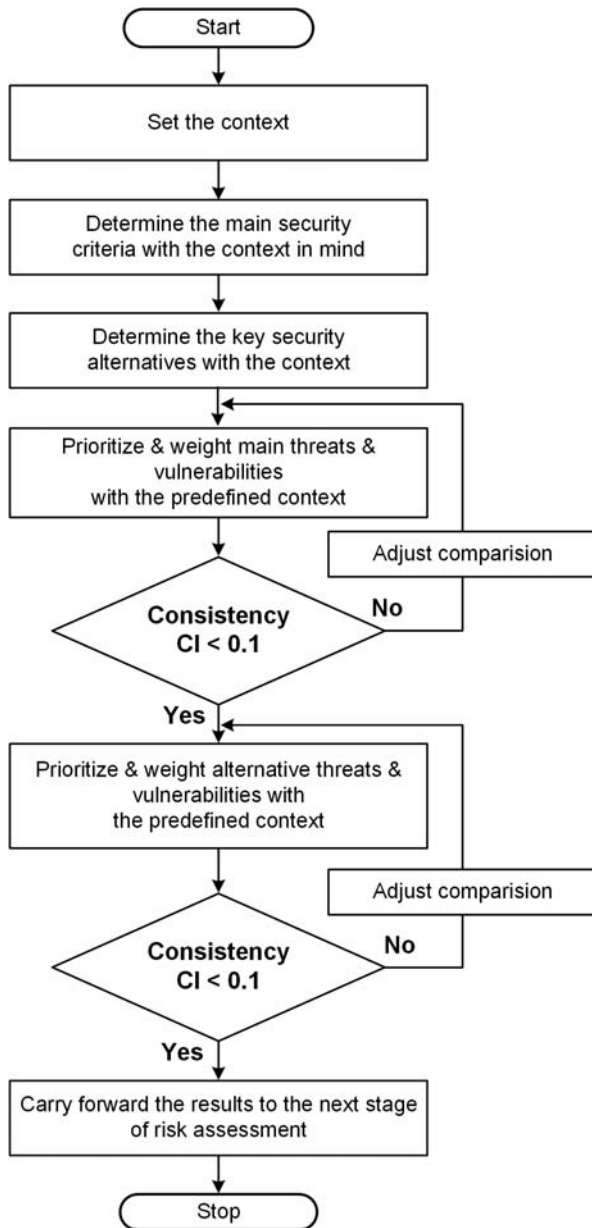


Figure 9. the Stepwise Process of Pairwise Comparison

Consistency Index (CI) and Consistency Ratio (CR)

The result gained from the pairwise comparison matrix A yields its maximum, $\lambda_{\max}$ (eigenvalue), and this is equal to 'n' only if the matrix is consistent.

Therefore, the consistency index, CI can be calculated with the given $\lambda_{\max}$;

$$CI(A) = \frac{(\lambda_{\max}) - n}{(n - 1)}$$

The next step is to compute the Consistency Ratio (CR) to quantify how consistent human perception have been relative to large samples of purely random figures.

With the given matrix (order of n), CR can be attained dividing CI by a Random Index (RI) which is a numerical estimation of the average CI obtained from a group of randomly generated matrices of size n.

$$CR = \frac{CI}{RI}$$

where the random index, RI, is a calculated random pairwise comparison for different size simulated matrices and is given in Table 1.

Table 1. Saaty's Random Index

n	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.5247	0.8816	1.1086	1.2479	1.3417	1.4057	1.4499	1.4854

According to Saaty, a $CR < 0.1$ implies consistency, while if it is not less than 10%, the judgments need to be revised [8].

De-Fuzzification of Risk Attributes Using Fuzzy Logic

In this method, the fuzzy logic reasoning is used to enhance the security risk assessment process while considering various uncertainties in the human interpretation of the likelihood of a security event and the severity of the same. Since these two parameters cannot be measured numerically, the associated risks are difficult to measure in crisp terms [11].

The fuzzy logic theory provides a new approach to dealing with incomplete, vague, and imprecise circumstances. The underlying architecture of a Fuzzy Expert System (FES) use for information de-fuzzification is depicted in Figure 10.

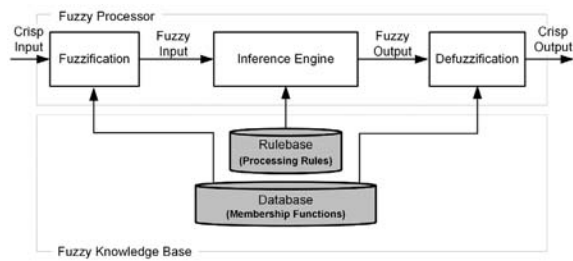


Figure 10. The Basic Architecture of a Fuzzy Expert System

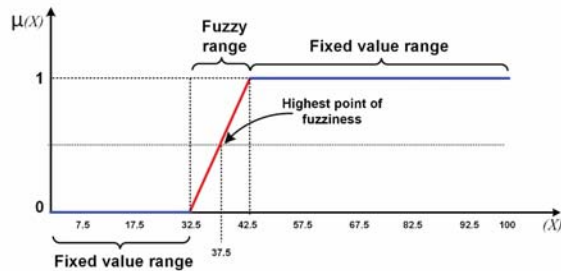


Figure 11. Membership Function Representing Imprecision in Crisp Value Variations

Conventionally, threats are categorised based on severity, for example, into one of the five level Figure 12(a). If the assessed degree is assumed to be between 62.5 and 87.5, it is deemed as 'High' even though the assessment results yield numeric value 65. In fuzzy logic theory, the assessment performs in shades of colors where the threat level is treated as a series of overlapping ranges as shown in Figure 12(b).

For example, in the fuzzy logic method, the assessed threat level 65 means 20% of 'Medium' and 80% of 'High' risk level. This classification provides a more accurate figure compared to the reading 100% of 'High' risk level with the conventional analysis method.

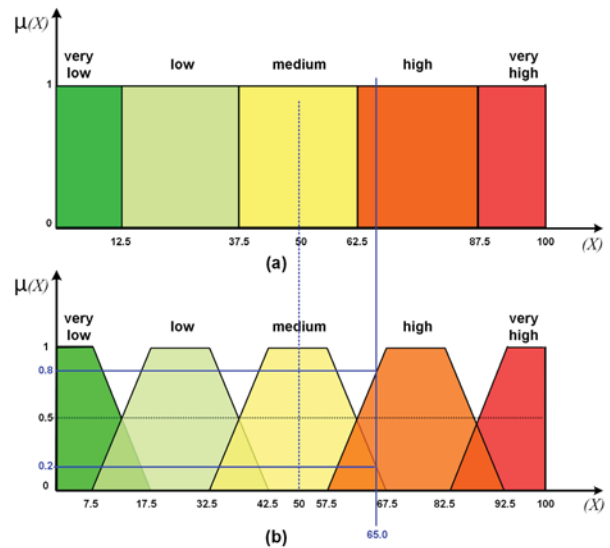


Figure 12. Conventional vs Fuzzy Logic Method

The Threat level- Likelihood matrix can be interpreted and transformed into membership functions (MF) as depicted in Figure 13.

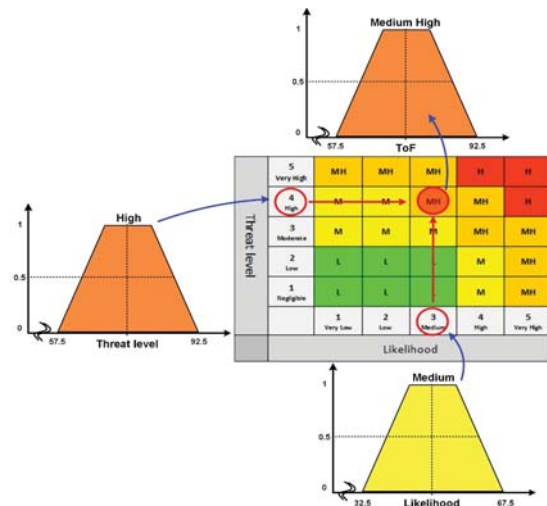


Figure 13. Risk Matrix Interpretation of Fuzzy Rule

Since evaluations are made on the testing of all security risk attribute rules, they must be consolidated in some way to conclude. Aggregation is the process used to characterise the outputs of each state and combined into a single fuzzy set.

The weighted averaged output method is used here for aggregation, and it can only be used in symmetrical output membership functions (Figure 14). Since decisions are made on assessing of all security

risk attributes, they must be combined using an aggregation process that used to characterise the outputs of each rule and combined into a single fuzzy set (Figure 14).

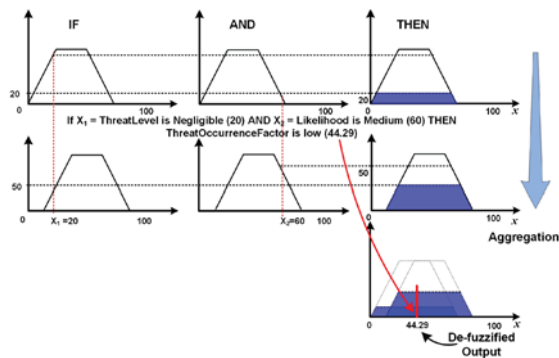


Figure 14. Fuzzy Inference Process

The proposed risk assessment methodology is modeled and simulated for various threats, likelihood and vulnerability attributes (Table 2) and the simulation is depicted in Figure 15.

Table 2. Simulation Test Data

Attribute	Linguistic Assessment	Matrix Reading	Fuzzy input
Threat level	High	4	82
Likelihood	Medium	3	65
Vulnerability	High	3	77
Asset value	<100K	1	10
Annual rate of occurrence	Very low	1	8

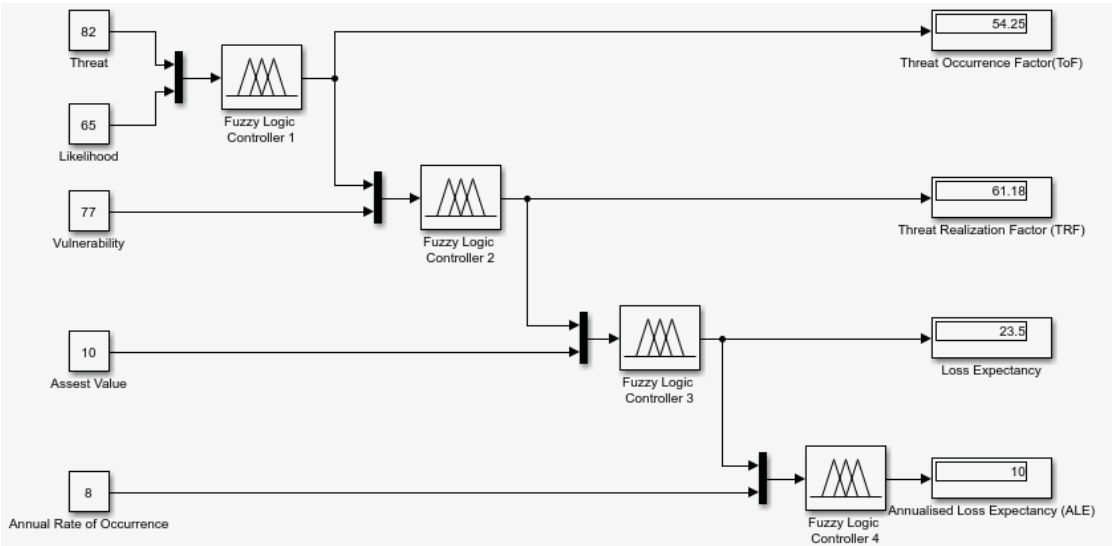


Figure 15. Fuzzy Logic Simulation (with MATLAB Simulink)

The test data for the given threat and likelihood scenario is fed into the Fuzzy Logic Controller (FLC1) and the simulator yields assorted numeric figures for the Threat Occurrence Factor (TOF) in terms of likelihood and threat level (Figure 16).

Similarly, the TOF output is fed to FLC2 along with the vulnerability evaluation for the chosen system, so that the Threat Realisation Factor (TRF) is calculated. The simulation result is shown in Figure 17.

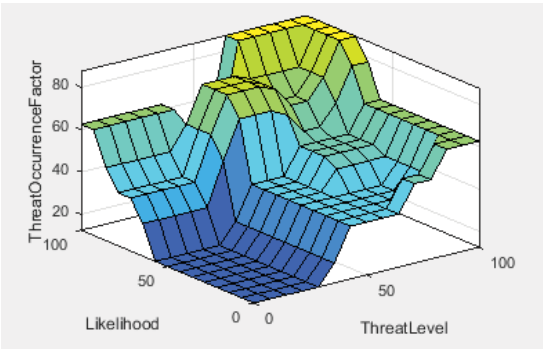


Figure 16. Threat Occurrence Factor

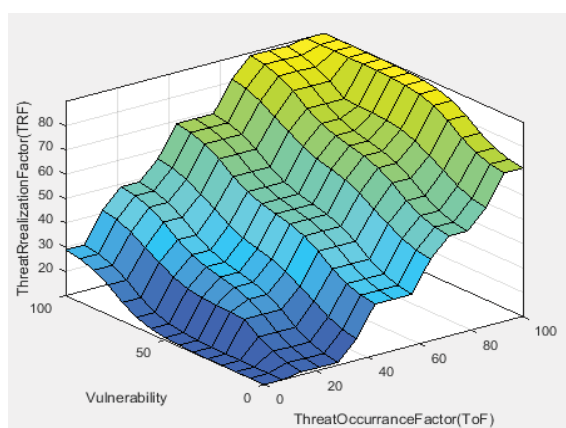


Figure 17. Threat Realisation Factor

In the information security, an asset is a system, resource, process, product, or people that have some value to a business and must, therefore, be protected. The assets are valued, and loss of information to the organisation is assessed with the aim of providing a clear account of the cost involved in protecting information. The calculated asset value (which includes installation, maintenance, and insurance cost as well) is then fed into FLC3 along with the TRF calculated from the FLC2. The resultant is the Loss Expectancy of the subject system and depicted in Figure 18.

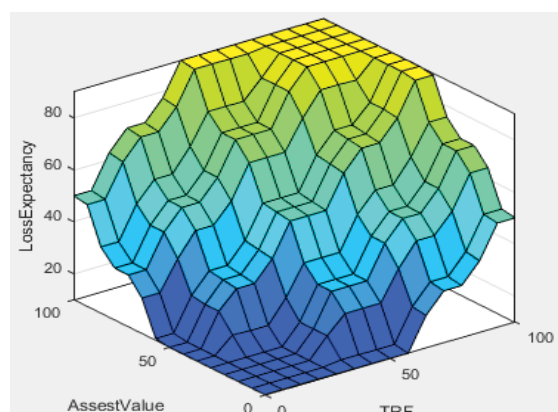


Figure 18. Loss Expectancy

Finally, the Annual Rate of Occurrence (ARO) is calculated using probability theory or by statistical analysis. This number is very much dependent on the security controls that are in place to protect the assets and require insight of security professionals and system auditors. The ARO along with the loss expectancy is fed to FLC4 to obtain annualised loss

expectancy of the system due to a given security incident and depicted in Figure 19.

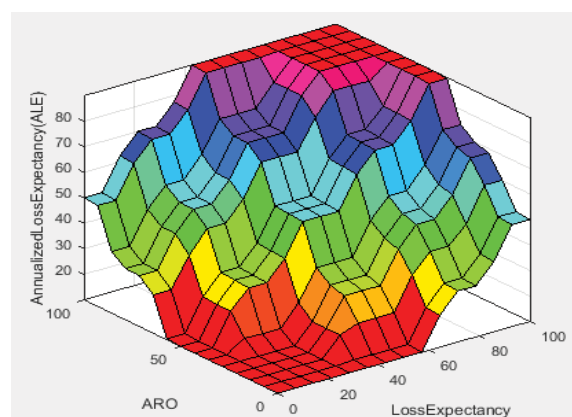


Figure 19. Annualised Loss Expectancy

Conclusion

The work described herein is a part of a research project to develop a cyber risk assessment methodology for civil aviation systems. The conventional risk assessment techniques currently being used for the purpose are simple and lead to inconsistent results. We propose a system engineering approach to the risk assessment framework intended to accommodate and understand cyber and cyber-physical attacks, covering all areas of civil aviation sphere. The procedure described in this paper can be used to prioritise critical scenarios using AHP so that it improves the consistency of decision making. Also, this methodology utilises the fuzzy inference engine to accomplish the best matching risk attributes so that most effective remediation measures can be employed to protect aviation systems from evolving threats and plausible system vulnerabilities.

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Acronyms

ACARS	Aircraft Communications, Addressing and Reporting System
ADS-B	Automatic Dependent Surveillance - Broadcast
AHM	Aircraft Health Monitoring
ASMGCS	Advanced-Surface Movement Guidance and Control System
ATN	Aeronautical Telecommunication Network
ATSP	Air Traffic Service Provider
AWOS	Automated Weather Observing System
EFB	Electronic Flight Bag
GBAS	Ground Based Augmentation System
GPSRAIM	GPS Receiver Autonomous Integrity Monitoring (RAIM)
ICS	Industrial Control System
LSAP	Loadable Software Airplane Parts
SQL	Structured Query Language (database)
SSL	Secure Sockets Layer
Stuxnet	A type of malicious computer worm
TSL	Transport Layer Security
VDAR	Virtual Digital Aircraft condition monitoring system Recorder
VPN	Virtual Private Network
XSS	Cross Site Scripting

Disclaimer

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